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DS Homework Set – 3

1) Write a C program to store n elements in an array using dynamic memory allocation and find largest number.

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int n, i, largest;
    int *arr;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL)
    {
        printf("Memory allocation failed");
        return 1;
    }
    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    largest = arr[0];
    for (i = 1; i < n; i++)
    {
        if (arr[i] > largest)
            largest = arr[i];
    }
    printf("Largest number = %d\n", largest);
    free(arr);
    return 0;
}
```

Output :

```
Enter the number of elements: 5
Enter 5 elements:
10 25 7 42 18
Largest number = 42
```

2) Print an array in reverse order using pointers.

```
#include <stdio.h>
int main()
{
    int n, i;
    int arr[100];
    int *ptr;
```

```
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    ptr = &arr[n - 1];
    printf("Array in reverse order:\n");
    for (i = n - 1; i >= 0; i--)
    {
        printf("%d ", *ptr);
        ptr--;
    }
    return 0;
}
```

Output :

```
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
Array in reverse order:
50 40 30 20 10
```

3) Write a C program to store information of Students using structure. Sort them by Name.

```
#include <stdio.h>
#include <string.h>
struct Student
{
    char name[50];
    int roll_no;
    float marks;
};
int main()
{
    int n, i, j;
    struct Student s[50], temp;
    printf("Enter number of students: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++)
    {
        printf("\nEnter details of student %d:\n", i + 1);
        printf("Name: ");
        scanf(" %[^\n]", s[i].name);
        printf("Roll No: ");
        scanf("%d", &s[i].roll_no);
        printf("Marks: ");
        scanf("%f", &s[i].marks);
    }
    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
```

```
        if (strcmp(s[i].name, s[j].name) > 0)
        {
          temp = s[i];
          s[i] = s[j];
          s[j] = temp;
        }
      }
    }
    printf("\nStudents sorted by name:\n");
    for (i = 0; i < n; i++)
    {
      printf("\nName: %s", s[i].name);
      printf("\nRoll No: %d", s[i].roll_no);
      printf("\nMarks: %.2f\n", s[i].marks);
    }
    return 0;
}
```

Output :

```
Enter number of students: 3

Enter details of student 1:
Name: Ravi
Roll No: 101
Marks: 85.5

Enter details of student 2:
Name: Arun
Roll No: 102
Marks: 91.0

Enter details of student 3:
Name: Karthik
Roll No: 103
Marks: 78.5

Students sorted by name:

Name: Arun
Roll No: 102
Marks: 91.00

Name: Karthik
Roll No: 103
Marks: 78.50

Name: Ravi
Roll No: 101
Marks: 85.50
```

4) Write a C program to add 'n' distances given in each-feet system using structure.

```
#include <stdio.h>
struct Distance
{
   int feet;
   int inches;
};
int main()
{
   int n, i;
   struct Distance d[50], sum = {0, 0};
   printf("Enter number of distances: ");
   scanf("%d", &n);
   for (i = 0; i < n; i++)
   {
      printf("\nEnter distance %d:\n", i + 1);
      printf("Feet: ");
      scanf("%d", &d[i].feet);
      printf("Inches: ");
      scanf("%d", &d[i].inches);
      sum.feet += d[i].feet;
      sum.inches += d[i].inches;
   }
   sum.feet += sum.inches / 12;
   sum.inches = sum.inches % 12;
   printf("\nTotal Distance = %d feet %d inches\n",
        sum.feet, sum.inches);
   return 0;
}
```

Output :

```
Enter number of distances: 3

Enter distance 1:
Feet: 5
Inches: 8

Enter distance 2:
Feet: 3
Inches: 10

Enter distance 3:
Feet: 4
Inches: 6

Total Distance = 14 feet 0 inches
```

5) Write a C program to add two complex numbers by passing structure to a function.

```
#include <stdio.h>
struct Complex
```

```
{
   float real;
   float imag;
};
struct Complex add(struct Complex c1, struct Complex c2)
{
   struct Complex result;
   result.real = c1.real + c2.real;
   result.imag = c1.imag + c2.imag;
   return result;
}
int main()
{
   struct Complex c1, c2, result;
   printf("Enter real and imaginary parts of first complex number: ");
   scanf("%f %f", &c1.real, &c1.imag);
   printf("Enter real and imaginary parts of second complex number: ");
   scanf("%f %f", &c2.real, &c2.imag);
   result = add(c1, c2);
   printf("Sum = %.2f + %.2fi\n",
        result.real, result.imag);
   return 0;
}
```

Output ;

```
Enter real and imaginary parts of first complex number: 3.5 2.5
Enter real and imaginary parts of second complex number: 4.5 1.5
Sum = 8.00 + 4.00i
```

6) Write a C program to medical inventory management system using structure.

```
#include <stdio.h>
struct Medicine
{
   int id;
   char name[50];
   int quantity;
   float price;
};
int main()
{
   int n, i;
   struct Medicine m[50];
   printf("Enter number of medicines: ");
   scanf("%d", &n);
   for (i = 0; i < n; i++)
   {
      printf("\nEnter details of Medicine %d:\n", i + 1);
      printf("Medicine ID: ");
      scanf("%d", &m[i].id);
      printf("Medicine Name: ");
      scanf(" %[^\n]", m[i].name);
```

```
        printf("Quantity: ");
        scanf("%d", &m[i].quantity);
        printf("Price: ");
        scanf("%f", &m[i].price);
    }
    printf("\n--- Medical Inventory ---\n");
    for (i = 0; i < n; i++)
    {
        printf("\nMedicine ID : %d", m[i].id);
        printf("\nName       : %s", m[i].name);
        printf("\nQuantity   : %d", m[i].quantity);
        printf("\nPrice      : %.2f\n", m[i].price);
    }
    return 0;
}
```

Output :

```
Enter number of medicines: 3

Enter details of Medicine 1:
Medicine ID: 101
Medicine Name: Paracetamol
Quantity: 50
Price: 25.50

Enter details of Medicine 2:
Medicine ID: 102
Medicine Name: Amoxicillin
Quantity: 30
Price: 45.75

Enter details of Medicine 3:
Medicine ID: 103
Medicine Name: Cetirizine
Quantity: 40
Price: 15.25

--- Medical Inventory ---

Medicine ID : 101
Name       : Paracetamol
Quantity   : 50
Price      : 25.50

Medicine ID : 102
Name       : Amoxicillin
Quantity   : 30
Price      : 45.75

Medicine ID : 103
Name       : Cetirizine
Quantity   : 40
```

Price : 15.25

7) Write a C program for decimal to binary, octal and hexadecimal conversion.

```
#include <stdio.h>
void binary(int n)
{
   int a[32], i = 0;
   if (n == 0)
   {
      printf("0");
      return;
   }
   while (n > 0)
   {
      a[i] = n % 2;
      n = n / 2;
      i++;
   }
   while (i > 0)
      printf("%d", a[--i]);
}
void octal(int n)
{
   int a[32], i = 0;
   if (n == 0)
   {
      printf("0");
      return;
   }
   while (n > 0)
   {
      a[i] = n % 8;
      n = n / 8;
      i++;
   }
   while (i > 0)
      printf("%d", a[--i]);
}
void hexadecimal(int n)
{
   int a[32], i = 0, rem;
   if (n == 0)
   {
      printf("0");
      return;
   }
   while (n > 0)
   {
      a[i] = n % 16;
      n = n / 16;
      i++;
   }
```

```
    while (i > 0)
    {
        rem = a[--i];
        if (rem < 10)
            printf("%d", rem);
        else
            printf("%c", 'A' + rem - 10);
    }
}
int main()
{
    int n;
    printf("Enter a decimal number: ");
    scanf("%d", &n);
    printf("Binary     : ");
    binary(n);
    printf("\nOctal      : ");
    octal(n);
    printf("\nHexadecimal : ");
    hexadecimal(n);
    return 0;
}
```

Output :

```
Enter a decimal number: 25
Binary     : 11001
Octal      : 31
Hexadecimal : 19
```

8) Write a C program to calculate difference between two time period.

```
#include <stdio.h>
struct Time
{
    int hours;
    int minutes;
    int seconds;
};
int main()
{
    struct Time t1, t2, diff;
    int time1, time2, difference;
    printf("Enter first time (hours minutes seconds): ");
    scanf("%d %d %d", &t1.hours, &t1.minutes, &t1.seconds);
    printf("Enter second time (hours minutes seconds): ");
    scanf("%d %d %d", &t2.hours, &t2.minutes, &t2.seconds);
    time1 = t1.hours * 3600 + t1.minutes * 60 + t1.seconds;
    time2 = t2.hours * 3600 + t2.minutes * 60 + t2.seconds;
    if (time1 > time2)
        difference = time1 - time2;
    else
        difference = time2 - time1;
```

```
    diff.hours = difference / 3600;
    difference = difference % 3600;
    diff.minutes = difference / 60;
    diff.seconds = difference % 60;
    printf("Difference = %d hours %d minutes %d seconds\n",
           diff.hours, diff.minutes, diff.seconds);
    return 0;
}
```

Output :

```
Enter first time (hours minutes seconds): 10 30 45
Enter second time (hours minutes seconds): 8 15 20
Difference = 2 hours 15 minutes 25 seconds
```

9) Write a C program to create an employee management system using structures.

```
#include <stdio.h>
struct Employee
{
    int id;
    char name[50];
    float salary;
};
int main()
{
    int n, i;
    struct Employee e[50];
    printf("Enter number of employees: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++)
    {
        printf("\nEnter details of Employee %d:\n", i + 1);
        printf("Employee ID: ");
        scanf("%d", &e[i].id);
        printf("Employee Name: ");
        scanf(" %[^\n]", e[i].name);
        printf("Salary: ");
        scanf("%f", &e[i].salary);
    }
    printf("\n--- Employee Details ---\n");
    for (i = 0; i < n; i++)
    {
        printf("\nEmployee ID   : %d", e[i].id);
        printf("\nEmployee Name : %s", e[i].name);
        printf("\nSalary        : %.2f\n", e[i].salary);
    }
    return 0;
}
```

Output :

```
Enter number of employees: 3
```

```
Enter details of Employee 1:
Employee ID: 101
Employee Name: Arun Kumar
Salary: 35000

Enter details of Employee 2:
Employee ID: 102
Employee Name: Ravi Kumar
Salary: 42000

Enter details of Employee 3:
Employee ID: 103
Employee Name: Karthik Raj
Salary: 38000

--- Employee Details ---

Employee ID   : 101
Employee Name : Arun Kumar
Salary        : 35000.00

Employee ID   : 102
Employee Name : Ravi Kumar
Salary        : 42000.00

Employee ID   : 103
Employee Name : Karthik Raj
Salary        : 38000.00
```

10) Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

```c
#include <stdio.h>
float rectangleArea(float l, float b)
{
   return l * b;
}
float rectanglePerimeter(float l, float b)
{
   return 2 * (l + b);
}
float triangleArea(float b, float h)
{
   return 0.5 * b * h;
}
float trianglePerimeter(float a, float b, float c)
{
   return a + b + c;
}
int main()
{
   float l, b, h;
   float a, side2, side3;
```

```c
    float (*rectArea)(float, float);
    float (*rectPerimeter)(float, float);
    float (*triArea)(float, float);
    float (*triPerimeter)(float, float, float);
    printf("Enter length and breadth of rectangle: ");
    scanf("%f %f", &l, &b);
    rectArea = rectangleArea;
    rectPerimeter = rectanglePerimeter;
    printf("\nRectangle Area = %.2f", rectArea(l, b));
    printf("\nRectangle Perimeter = %.2f", rectPerimeter(l, b));
    printf("\n\nEnter base and height of triangle: ");
    scanf("%f %f", &b, &h);
    printf("Enter three sides of triangle: ");
    scanf("%f %f %f", &a, &side2, &side3);
    triArea = triangleArea;
    triPerimeter = trianglePerimeter;
    printf("\nTriangle Area = %.2f", triArea(b, h));
    printf("\nTriangle Perimeter = %.2f",
           triPerimeter(a, side2, side3));
    return 0;
}
```

Output :

```
Enter length and breadth of rectangle: 10 5

Rectangle Area = 50.00
Rectangle Perimeter = 30.00

Enter base and height of triangle: 6 4
Enter three sides of triangle: 5 6 7

Triangle Area = 12.00
Triangle Perimeter = 18.00
```